

Programming Exercises in R

Summary Statistics and Data Visualization

Task 1: Create a Numeric Vector and Draw a Bar Plot

Program

```
1 marks <- c(45,52,60,70,85,90,78,65)
2
3 print(marks)
4
5 barplot(marks,
6         col="skyblue",
7         main="Students Marks",
8         xlab="Student",
9         ylab="Marks")
```

Task 2: Find the Mean and Display it on a Bar Plot

Program

```
1 marks <- c(45,52,60,70,85,90,78,65)
2
3 m <- mean(marks)
4
5 barplot(marks,
6         col="lightgreen",
7         main="Mean of Marks")
8
9 abline(h=m,col="red",lwd=2)
10 text(6,m+2,paste("Mean =",round(m,2)),col="red")
```

Task 3: Find the Median and Plot the Data

Program

```
1 marks <- c(45,52,60,70,85,90,78,65)
2
3 med <- median(marks)
4
5 plot(marks,
6      type="b",
7      pch=19,
8      col="blue",
9      main="Median Illustration")
10
11 abline(h=med,col="red",lwd=2)
```

Task 4: Display Summary Statistics and Box Plot

Program

```
1 marks <- c(45,52,60,70,85,90,78,65)
2
3 summary(marks)
4
5 boxplot(marks,
6         col="orange",
7         main="Box Plot of Marks")
```

Task 5: Find the Range and Draw a Dot Chart

Program

```
1 marks <- c(45,52,60,70,85,90,78,65)
2
3 range(marks)
4
5 dotchart(marks,
6          pch=19,
7          col="blue",
8          main="Dot Chart")
```

Task 6: Find Variance and Draw a Histogram

Program

```
1 marks <- c(45,52,60,70,85,90,78,65)
2
3 var(marks)
4
5 hist(marks,
6       col="lightblue",
7       main="Histogram",
8       xlab="Marks")
```

Task 7: Find Standard Deviation and Plot a Line Graph

Program

```
1 marks <- c(45,52,60,70,85,90,78,65)
2
3 sd(marks)
4
5 plot(marks,
6       type="o",
7       pch=16,
8       col="darkgreen",
9       main="Line Plot of Marks")
```

Task 8: Find the Interquartile Range and Draw a Box Plot

Program

```
1 marks <- c(45,52,60,70,85,90,78,65)
2
3 IQR(marks)
4
5 boxplot(marks,
6          horizontal=TRUE,
7          col="pink",
8          main="IQR using Box Plot")
```

Task 9: Calculate Skewness and Plot Histogram with Density Curve

Program

```
1 library(moments)
2
3 marks <- c(45,52,60,70,85,90,78,65)
4
5 skewness(marks)
6
7 hist(marks,
8       probability=TRUE,
9       col="lightyellow",
10      main="Histogram with Density")
11
12 lines(density(marks),
13       col="red",
14       lwd=2)
```

Task 10: Calculate Kurtosis and Draw Histogram with Density Curve

Program

```
1 library(moments)
2
3 marks <- c(45,52,60,70,85,90,78,65)
4
5 kurtosis(marks)
6
7 hist(marks,
8       probability=TRUE,
9       col="lightgreen",
10      main="Distribution of Marks")
11
12 lines(density(marks),
13       col="blue",
14       lwd=2)
```

Probability and Statistics using R

Lab Session-3

Summary Statistics (Location, spread, shape and peakedness)

Course Code : MAT445

Date:30.07.2026

Time : 3:15PM - 4:15 PM

Programming Tasks

1. Create a vector of marks: 45, 56, 67, 72, 81, 90, 76, 68 and find the mean.
2. Create a vector of salaries: 25000, 28000, 32000, 30000, 29000, 31000, 27000 and find the median.
3. Find the minimum, maximum and range of the data:

12, 15, 18, 20, 24, 30, 35, 40

4. Compute the variance and standard deviation for the data:

14, 18, 20, 22, 24, 28, 30

5. Find the first quartile, median and third quartile of:

5, 8, 10, 12, 15, 18, 20, 22, 25

6. Find the interquartile range (IQR) for:

16, 18, 21, 23, 25, 28, 30, 35, 40

7. Display the complete summary statistics of:

55, 62, 70, 68, 75, 80, 90, 85

8. Install the `moments` package and calculate the skewness of the following data:

10, 12, 15, 18, 22, 40, 45

9. Calculate the kurtosis of the following observations:

5, 7, 9, 10, 12, 15, 20, 25, 50

10. Write an R program to calculate Mean, Median, Variance, Standard Deviation and IQR of a given dataset using a single program.

Answers

1. Mean of Marks

```
1 marks <- c(45,56,67,72,81,90,76,68)
2
3 mean(marks)
```

2. Median of Salaries

```
1 salary <- c(25000,28000,32000,
2             30000,29000,31000,27000)
3
4 median(salary)
```

3. Minimum, Maximum and Range

```
1 x <- c(12,15,18,20,24,30,35,40)
2
3 min(x)
4 max(x)
5 range(x)
```

4. Variance and Standard Deviation

```
1 x <- c(14,18,20,22,24,28,30)
2
3 var(x)
4 sd(x)
```

5. Quartiles

```
1 x <- c(5,8,10,12,15,18,20,22,25)
2
3 quantile(x)
```

6. Interquartile Range

```
1 x <- c(16,18,21,23,25,28,30,35,40)
2
3 IQR(x)
```

7. Summary Statistics

```
1 x <- c(55,62,70,68,75,80,90,85)
2
3 summary(x)
```

8. Skewness

```
1 install.packages("moments")
2
3 library(moments)
4
5 x <- c(10,12,15,18,22,40,45)
6
7 skewness(x)
```

9. Kurtosis

```
1 library(moments)
2
3 x <- c(5,7,9,10,12,15,20,25,50)
4
5 kurtosis(x)
```

10. Complete Summary Statistics Program

```
1 library(moments)
2
3 data <- c(25,28,30,35,40,
4           42,45,48,50,55)
5
6 cat("Mean =", mean(data), "\n")
7
8 cat("Median =", median(data), "\n")
9
10 cat("Variance =", var(data), "\n")
11
12 cat("Standard Deviation =", sd(data), "\n")
13
14 cat("IQR =", IQR(data), "\n")
15
16 cat("Skewness =", skewness(data), "\n")
17
18 cat("Kurtosis =", kurtosis(data), "\n")
19
20 summary(data)
```